

DISCRETE CHOICE WITH GENERALIZED SOCIAL INTERACTIONS

OSCAR VOLPE

Department of Economics, Harvard University

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This paper studies social interactions in discrete choice models in which individuals can respond differently to different network members, conforming to some while distinguishing from others. In this generalized framework, I show how to use within-network variation in individual choices to draw inference about differences in endogenous interaction effects, while accounting for unobserved contextual effects. I also show how to construct internal instruments to correct for measurement error bias, a key source of endogeneity in network models with incomplete information. I apply these results to data from two empirical settings: classroom peer effects in Tennessee primary schools and spillovers in deworming treatment uptake in Kenya. In both applications, I find that differences in social interaction effects, with individuals conforming more strongly to similar peers, play a key role in shaping people's choices, with implications for the targeting of policy interventions.

KEYWORDS: social interaction effects, discrete choice, strategic complements and substitutes, contextual effects, measurement error bias, partial identification.

1. INTRODUCTION

The role of social interactions in shaping individual behavior is an active research topic in economics. Most empirical work has focused on settings with uniform positive interactions, where all agents conform in the same way to the average action of others in their network. Meanwhile, less attention has been given to settings with nonuniform or mixed-sign interactions, where agents are affected differently by different network members—conforming to some while distinguishing from others. These richer forms of interaction can be harder to evaluate empirically, yet they are important for understanding behavior in many economic settings, where social identity shapes how people's choices depend on those around them.

Oscar Volpe: ovolpe@fas.harvard.edu

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In this paper, I develop empirical tools to study discrete choice models with *generalized* social interactions, where individuals can experience both positive and negative spillovers. I build an equilibrium framework that extends the classical model of [Brock and Durlauf \(2001\)](#) by allowing the sign and magnitude of interaction effects to vary among agents in a network. Under this framework, I show how to recover key economic quantities from individual choice data, and I construct an estimator for endogenous interaction effects that can be readily applied to standard datasets. I then apply these methods to data from two large-scale education experiments to draw inference about the role of nonuniform interaction effects.

In Section 2, I begin by modeling a network of agents who make binary choices subject to social interaction effects and private payoffs. As in [Brock and Durlauf \(2001\)](#), I assume agents act with incomplete information, responding to expected rather than realized choices of others. Yet, unlike previous work, I allow interaction effects to vary among agents and to take negative values. To achieve this, I extend the model to allow for a matrix of interaction effects $\mathbf{J} = [J_{k\ell}]_{k,\ell}$, where $J_{k\ell}$ measures how agents of type k respond to agents of type ℓ .

By imposing no restrictions on $\mathbf{J}$, I show that the model can be used to study a range of economic decision problems across different network structures, featuring both conformity and differentiation. Under this generalized setting, I establish conditions for the equilibrium to be unique and locally stable, characterizing how these properties depend on the signs and magnitudes of the interaction effects.¹ I also show that, when multiple equilibria exist, negative interactions can create welfare tradeoffs, in which no equilibrium is broadly favorable.

Next, in Section 3, I turn to the question of identification. That is, given data on individual choices across multiple networks, what can be learned about the role of social interactions? The identification of discrete choice network models with incomplete information has been studied extensively by [Brock and Durlauf \(2001, 2007\)](#) and more recently by [Bhattacharya et al. \(2024\)](#). [Graham \(2015\)](#), [de Paula \(2017\)](#), and [Kline and Tamer \(2020\)](#) give surveys of the literature. My analysis builds on this work by tackling two key identification challenges.

First, I address the issue of unobserved network effects—contextual factors that obscure the effect of social interactions on group-level outcomes. For example, from data on student achievement, it is hard to isolate the role of peer effects from unobserved class characteristics like teacher quality. Much of the applied literature rules out or places strong parametric restrictions on these unobservables.² However, this risks introducing omitted variable bias.

¹While these properties are well-studied for uniform, positive interactions, they do not readily transfer to cases with negative interactions; see [Jackson and Zenou \(2015\)](#) for discussion. My analysis contributes by showing how key results associated with positive interactions extend to a broader set of models involving negative interactions.

²For example, [Brock and Durlauf \(2001, 2007\)](#) assume that the contextual effect is a constant linear function of observed variables, whereas [Bhattacharya et al. \(2024\)](#) assume that it has a specific linear factor structure. Others,

To handle this issue, I propose a partial identification strategy to recover key features of $\mathbf{J}$ without imposing any restrictions on unobserved network effects. This method exploits a panel structure inherent to the model, where agents in the same network interact in varying ways. Since these agents are exposed to the same contextual factors, I can control for network unobservables by comparing outcomes of different agents in the same network. This method allows me to draw inference about systematic variation in social interaction effects.

Second, I address a core issue in network models with incomplete information: agents' expectations are not observed by the researcher. Instead, a researcher only observes agents' realized choices in finite networks. Since the average choice in the network converges to an expectation as the network size grows large, previous work has treated this issue as an estimation problem rather than a threat to identification.³ I argue that this approach is inadvisable, since, for any finite network, the average action is endogenous in the model due to measurement error. I show that, by failing to account for this endogeneity, standard estimators generally produce highly biased estimates, even as the number of networks tends to infinity.

I approach this issue as a classical measurement error problem, noting that realized average choices are noisy proxies for true expectations and are therefore endogenous. To correct for endogeneity, I construct internal instruments: randomly partitioning networks into two parts, and using the average action in one part as an instrument for the average action in the other. Under the model's assumptions, this method yields an IV estimator for social interaction effects that is consistent as the number of networks grows, even when network sizes remain fixed. I demonstrate that this estimator performs well in Monte Carlo simulations.

Finally, in Section 4, I apply these results to data from two empirical settings. First, I analyze Project STAR—a large-scale education experiment in Tennessee that randomly assigned students to classrooms—to study how peer effects vary by gender. I find that boys and girls are much more likely to conform to peers of their own gender than to peers of the other gender. Next, I use data from the Primary School Deworming Project, previously studied by Miguel and Kremer (2004), to estimate how spillovers in treatment uptake vary by age and gender. Again, I find that students prefer to conform to similar peers. Moreover, this own-group conformity is strongest among younger students. In both applications, I find that the generalized interactions model uncovers systematic differences in interaction effects that standard frameworks would overlook. Accounting for these differences provides a richer understanding of how social interactions shape the effects of targeted policies in networks.

including Hoxby (2000), propose using a panel to difference-out the fixed effects between time periods. However, this approach requires both access to panel data and the assumption that contextual effects are invariant over time.

³Previously, estimation has relied on double asymptotic arguments, requiring both the number and size of the networks to tend to infinity; e.g., see Lee et al. (2014), Guerra and Mohnen (2022), and Bhattacharya et al. (2024).

2. MODEL WITH GENERALIZED INTERACTION EFFECTS

In this section, I introduce a binary choice model with social interactions that extends the classical model of [Brock and Durlauf \(2001\)](#) by allowing the sign and strength of interaction effects to vary across individuals. Unlike in the classical model, the analysis is nonparametric in the distribution of idiosyncratic preferences. Therefore, the equilibrium properties and subsequent identification results are robust to a wide range of functional form assumptions.

2.1. Agent Preferences

Consider a network of agents indexed by i , where an agent belongs to one of K types. These types can represent social identities, such as gender, class, race, or political partisanship, or they can represent organizational roles within the network, such as workers and supervisors, suppliers and manufacturers, patients and clinicians, or villagers and community leaders.

Each agent chooses a binary action $\omega_i \in \{0, 1\}$ at a common time. Let $\bar{\omega}^k$ be the average action among agents of type k , and let $\bar{\omega}_{-i}^k$ be the average action among agents of type k excluding i . The individual utility from choosing an action ω_i for any agent i of type k is:

$$U_i(\omega_i|k) = v_k(\omega_i) + J_{kk}\omega_i E_i(\bar{\omega}_{-i}^k) + \sum_{\ell \neq k} J_{k\ell}\omega_i E_i(\bar{\omega}^\ell) + \epsilon_i(\omega_i). \quad (1)$$

In this utility function, $v_k(\omega_i)$ is the deterministic private payoff, which can depend on the agent's type k , and $\epsilon_i(\omega_i)$ is an idiosyncratic payoff, which can vary across individuals. The terms $E_i(\bar{\omega}_{-i}^k)$ and $E_i(\bar{\omega}^\ell)$ are agent i 's subjective beliefs about $\bar{\omega}_{-i}^k$ and $\bar{\omega}^\ell$, respectively.⁴

The parameter $J_{k\ell}$ is the interaction effect, measuring how agents of type k are influenced by the expected average choice of agents of type ℓ . These terms can be collected in a matrix:

$$\mathbf{J} = \begin{bmatrix} J_{11} & J_{12} & \cdots & J_{1K} \\ J_{21} & J_{22} & \cdots & J_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ J_{K1} & J_{K2} & \cdots & J_{KK} \end{bmatrix}. \quad (2)$$

When analyzing model behavior, I refer to $\mathbf{J}$ as the *interaction matrix*. Its entries characterize the direction, sign, and magnitude of social influence among different types of agents.⁵

Since choices are binary, the private payoffs can, without loss of generality, be written as linear functions of the choice: $v_k(\omega_i) = h_k\omega_i + \eta_k$ and $\epsilon_i(\omega_i) = \varepsilon_i\omega_i + \xi_i$. Here, h_k is the de-

⁴ Another common way to model spillovers is through a quadratic conformity effect, $-\frac{1}{2}J_{k\ell}[\omega_i - E_i(\bar{\omega}^\ell)]^2$, as studied by [Bernheim \(1994\)](#). This parameterization is equivalent in that it leads agents to make the same choices.

⁵ $\mathbf{J}$ can be interpreted as the adjacency matrix of a directed graph, where nodes represent different agent types.

terministic payoff from selecting $\omega_i = 1$, and ε_i is the idiosyncratic taste for this action. The tastes are assumed independent across agents, and they can have type-specific distributions:

$$P(\varepsilon_i \leq z | k) = F_{\varepsilon|k}(z), \quad \text{for } k = 1, \dots, K, \quad (3)$$

where $F_{\varepsilon|k}$ is strictly increasing, smooth, and symmetric about zero. No additional parametric assumptions are maintained throughout the analysis. Therefore, the framework accommodates a variety of common empirical specifications, such as logistic and Gaussian errors.

2.2. Economic Examples of Generalized Interactions

The classical model of social interactions for binary outcomes (Brock and Durlauf, 2001) imposes two key restrictions on the interaction effects. First, it assumes uniformity ($J_{k\ell} = J$ for all k and ℓ), so all agents influence each other in the same way, regardless of their types. Second, it assumes that all interaction effects are nonnegative ($J_{k\ell} \geq 0$ for all k and ℓ), so each agent's payoff weakly rises when others take the same action. This restriction rules out strategic substitutability, where agents seek to differentiate from certain network members.

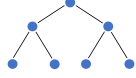
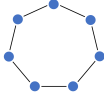
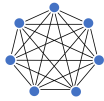
I relax the first restriction by allowing the interaction effects to vary among types and also by not imposing symmetry ($J_{k\ell} = J_{\ell k}$), so the influence of type k on type ℓ can differ from that of type ℓ on type k . This generalization makes the model well-suited to study a variety of economic decision problems across different network structures. Table 1, Panel A, presents four commonly studied network structures in economics, as well as the conditions on $\{J_{k\ell}\}_{k,\ell}$ that define each one. The first is a *tree*, used to study spillovers in organizational hierarchies and supply chains, as well as the spread of information or disease. The second is a *circle*, relevant for analyzing local interactions among close contacts (Ellison, 1993) and spatial competition between nearby firms (Salop, 1979). The third is a *bipartite* network, used to study two-sided markets linking workers and firms or buyers and sellers. The fourth is a *complete* network, in which all agents interact with one another, as in classical peer effects models. Each of these networks is nested under the generalized interactions framework.

I relax the second restriction by allowing for any combination of positive and negative interaction effects. This extension makes the model applicable to environments where agents seek to differentiate their behavior from that of others. Examples, shown in Table 1, Panel B, include *free-riding* (Bramoullé and Kranton, 2007), where agents contribute less to a public good when others contribute more; *social distinction* (Akerlof and Kranton, 2000), where agents differentiate their choices to reinforce identity; *segregation* (Schelling, 1971), where agents prefer actions that result in the exclusion of particular peers; and *polarization* (Boxell et al., 2024), where agents deviate from actions taken by those with opposing views. In each case, an agent's preference to distinguish her choice from those of others can be modeled as

a negative interaction effect. Thus, by allowing for a mix of positive and negative interaction effects, the generalized model accommodates any pattern of conformity and differentiation.

Relative to the classical model, the generalized interactions framework admits a new set of target parameters that provide insight into how social interactions vary among different individuals. For example, the difference $J_{k_1\ell} - J_{k_2\ell}$ measures how the influence of type ℓ differs between agents of types k_1 and k_2 . These terms can be used to construct an economic measure of polarization, $\delta_{k_1k_2} = J_{k_1k_1} + J_{k_2k_2} - J_{k_1k_2} - J_{k_2k_1}$, which quantifies the two types' combined preference for conforming to their own type rather than to the other type.

TABLE 1
COMMONLY STUDIED ECONOMIC NETWORKS AND INTERACTION EFFECTS

	Illustration ($K = 7$)	Structural Representation	Economic Examples
<i>Panel A: Network Structures</i>			
Tree		$\sum_{k < \ell} \mathbb{1}\{ J_{k\ell} + J_{\ell k} > 0\} = K - 1$ and all types connected	hierarchy, vertical production, contagion, info diffusion
Circle		$J_{k\ell} \neq 0$ for $\ell = k \pm 1 \pmod{K}$	local interactions, spatial competition
Bipartite Network		$J_{k\ell} \neq 0$ if $k \in A$ and $\ell \in B$ for disjoint $A, B \subseteq \{1, \dots, K\}$	labor market, product market, college admission/matching
Complete Network		$J_{k\ell} \neq 0$ for all k, ℓ	classroom peer effects, public goods settings, oligopoly
<i>Panel B: Modes of Social Influence</i>			
Strategic Complementarity	$\frac{\partial[U_i(1 k) - U_i(0 k)]}{\partial E_i(\bar{\omega}^\ell)} \geq 0$	$J_{k\ell} \geq 0$	conformity/imitation, social learning, technology adoption
Strategic Substitutability	$\frac{\partial[U_i(1 k) - U_i(0 k)]}{\partial E_i(\bar{\omega}^\ell)} \leq 0$	$J_{k\ell} \leq 0$	free-riding, social distinction, segregation, polarization

Notes. This table lists examples of network structures and spillovers nested by the generalized interactions model.

2.3. Equilibrium under Noncooperative Decisionmaking

My analysis focuses on pure strategy Bayesian Nash equilibria in which agents act noncooperatively, that is, without coordination. An agent of type k chooses $\omega_i = 1$ with probability:

$$P(\omega_i = 1|k) = F_{\varepsilon|k} \left(h_k + J_{kk} E_i(\bar{\omega}_{-i}^k) + \sum_{\ell \neq k} J_{k\ell} E_i(\bar{\omega}^\ell) \right). \quad (4)$$

Since ω_i takes values in the set $\{0, 1\}$, the expected action $E(\omega_i|k)$ also equals $P(\omega_i = 1|k)$.

Agents act with incomplete information. Specifically, an agent i knows her own idiosyncratic taste ε_i and the deterministic preference parameters $(h_k, \{J_{k\ell}\}_\ell, F_{\varepsilon|k})$ for each type, but not the idiosyncratic tastes ε_j of other agents $j \neq i$. This information structure can be interpreted in two ways. First, it may be that agents do not see everyone else's choice, either because actions are taken simultaneously, such as during a test, or because the network is too large to track individual actions, such as in a school or village. Under this interpretation, ε_i reflects the uncertainty that agents have about others' choices. A second interpretation of the information structure is that agents respond to social norms—the behavior of a “typical” agent of a given type—while ignoring idiosyncratic deviations from those norms. This interpretation is consistent with economic theories of social influence mechanisms ([Akerlof and Kranton, 2000](#)), where norms are understood to reflect shared values of a social group.

An equilibrium is characterized by the expected average choices that are consistent with individually optimal decisions. I assume that agents hold consistent beliefs, so that they correctly anticipate other agents' actions in expectation: $E_i(\omega_j|k) = E(\omega_j|k)$ for all $i \neq j$ and k . Letting m^k denote $E(\bar{\omega}^k)$, an equilibrium $m = (m^1, \dots, m^K)$ is defined as a solution to:

$$m^k = F_{\varepsilon|k} \left(h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell \right), \quad \text{for } k = 1, \dots, K. \quad (5)$$

Since $F_{\varepsilon|k}$ is continuous, Brouwer's fixed point theorem ensures that an equilibrium exists.

2.4. Equilibrium Uniqueness and Local Stability

I now examine two key equilibrium properties of the model: uniqueness and local stability. Uniqueness asserts that, for a fixed set of parameters, there is one solution to the equilibrium system. Local stability concerns whether iteration on agents' best responses converges to an equilibrium. Formally, m is *locally stable* if it is a limiting solution to the dynamical system $m_t^k = F_{\varepsilon|k}(h_k + \sum_{\ell=1}^K J_{k\ell} m_{t-1}^\ell)$ for $k = 1, \dots, K$, where the starting value m_0 lies within some sufficiently small neighborhood of m . Meanwhile, m is *unstable* if starting values m_0 arbitrarily close to m can produce an iterate m_t outside some fixed neighborhood of m .

Both properties are fundamental to comparative statics. If a locally stable equilibrium is perturbed slightly, then behavioral outcomes would return to that equilibrium. Meanwhile, if an equilibrium is unstable, then nearby outcomes would diverge from it. In settings with multiple equilibria, a locally stable equilibrium is one that a researcher observes in practice, while an unstable equilibrium acts as a tipping point between stable equilibrium outcomes.

2.4.1. Settings with a Unique Locally Stable Equilibrium

I begin by establishing conditions for the equilibrium to be unique and locally stable. Under the generalized interactions model, these conditions depend on both the network structure and the strength of social interactions relative to private preferences. I find that a unique and locally stable equilibrium exists when network ties are sparse, limiting behavioral feedback, or when interaction effects are weak, so private payoffs tend to dominate individual utility.

To obtain these conditions, I first derive the following property, which shows that, generically, multiple equilibria always include at least one unstable equilibrium. Hence, to establish uniqueness and local stability, it suffices to show that every equilibrium is locally stable.

PROPERTY 1: *For almost all distributions $\{F_{\varepsilon|k}\}_k$, the model has a finite, odd number of equilibria. If d_s of these equilibria are locally stable, then at least $d_s - 1$ must be unstable.*

To evaluate equilibrium stability, I examine the *Jacobian matrix* of the equilibrium system:

$$\mathbf{D}(m) = \beta(m)\mathbf{J}. \quad (6)$$

Here, $\beta(m)$ is a diagonal matrix whose k th entry is $f_{\varepsilon|k}(h_k + \sum_{\ell=1}^K J_{k\ell}m^\ell)$, the density of type- k agents at the decision threshold. Thus, the Jacobian weights each row of $\mathbf{J}$ by the responsiveness of the corresponding agent type. Fixing $\mathbf{J}$, choice probabilities respond more strongly to changes in others' behavior when a greater share of agents are near indifference.

Equilibrium stability is governed by the *spectral radius* of the Jacobian matrix, defined as the largest of its eigenvalues in absolute value: $\rho(\mathbf{D}(m)) = \max\{|\lambda| : \lambda \text{ eigenvalue of } \mathbf{D}(m)\}$. This object measures the overall strength of equilibrium feedback. A value below one implies the equilibrium is locally stable; a value above one implies the equilibrium is unstable.⁶

LEMMA 1: *If $\rho(\mathbf{D}(m)) < 1$, then m is locally stable; if $\rho(\mathbf{D}(m)) > 1$, then m is unstable.*

Together, these results establish that the equilibrium is unique and locally stable whenever $\rho(\mathbf{D}(m)) < 1$ at all equilibria m . The following examples illustrate two ways this can occur.

Example 1 (One-Way Interactions). Suppose social influence flows only in one direction, so that all feedback effects of the form $J_{kj_1}J_{j_1j_2}\cdots J_{j_Mk}$ equal zero. This structure arises in hierarchies or information networks, where influence flows from upstream to downstream

⁶The setting $\rho(\mathbf{D}(m)) = 1$ represents a knife-edge case that occurs with probability zero and has no particular economic relevance in the model. In such cases, equilibrium stability depends on higher-order derivatives of (5).

agents but never returns to its source. In this case, behavior cannot become self-reinforcing. The model implies $\rho(\mathbf{D}(m)) = 0$ for all m , so the equilibrium is unique and locally stable.

Example 2 (Weak Interactions). Suppose the interaction effects are weak relative to private payoffs. Specifically, for any type k , let $\sum_{\ell=1}^K |J_{k\ell}| < S_k$, where S_k is a threshold that depends on deterministic private payoffs and the distribution of idiosyncratic tastes.⁷ This condition places a bound on the overall amount of social influence exerted on each type. It implies that $\rho(\mathbf{D}(m)) < 1$ at every m , so the equilibrium must be unique and locally stable.

2.4.2. Settings with Multiple Locally Stable Equilibria

I next examine when the model has multiple locally stable equilibria. This question is well-understood in the classical model with uniformly positive interaction effects ($J_{k\ell} \geq 0$ for all k and ℓ). In this case, whenever social interactions are strong enough to generate an unstable equilibrium, the model contains multiple equilibria, at least two of which are locally stable.⁸

This result does not generally extend to settings with negative interactions. Indeed, models with negative interactions may have a unique but unstable equilibrium, such that agents fail to converge on a single set of choices. To see why, suppose that $J_{kk} < 0$, so agents of type k prefer to differentiate from their own type. When $-J_{kk}$ is sufficiently large, every equilibrium is unstable: if most of type k selects the high action, its members would rather switch to the low action, and vice versa. Similar instability can arise from conflicting cross-type interactions. For example, if $J_{k\ell} > 0$ and $J_{\ell k} < 0$, then type k seeks to conform to type ℓ , while type ℓ seeks to differentiate from type k . When these effects are sufficiently strong, each type's response can undermine the other's, ruling out any locally stable equilibrium.⁹

The contribution of my analysis is to pinpoint the source of global instability in models with negative interactions, and use this insight to classify which models can sustain locally stable equilibria, even with strong spillovers. The examples above suggest that global instability occurs when agents strongly wish to deviate from other members of their own type. In that case, agents never settle on a choice, as they are always dissatisfied with the outcome.

To rule out global instability, it is crucial that social interactions do not lead to self-contradictory preferences. Specifically, after accounting for all feedback effects $J_{kj_1} J_{j_1 j_2} \cdots J_{j_M k}$,

⁷In a model in which $f_{\varepsilon|k}$ is unimodal and $h_k = 0$ for all types k , the threshold S_k simplifies to $S_k = 1/f_{\varepsilon|k}(0)$.

⁸The microeconomic foundations for models of complementarity are largely developed by Donald Topkis (see Topkis (1998) for a summary), Vives (1990), and Milgrom and Roberts (1990). This work uses key results from lattice theory, e.g., Tarski's fixed point theorem, to study equilibrium behavior under uniform complementarities.

⁹This setting is strategically similar to a matching pennies game, which has no pure strategy Nash equilibrium. As another example, if $\mathbf{J}$ is symmetric and its eigenvalues have nonpositive real parts, then there is only one equilibrium, which is locally stable when $\rho(\mathbf{D}(m)) < 1$ and unstable when $\rho(\mathbf{D}(m)) > 1$; see Appendix Lemma A.3.

agents should not be strongly repelled by their own type. I formalize this condition below:

ASSUMPTION A: There is an invertible matrix $\mathbf{B}$ for which $\mathbf{B}\mathbf{D}(\cdot)\mathbf{B}^{-1} \geq 0$ entrywise.

Assumption A states that, under a change of basis, $\mathbf{D}$ can be transformed into a nonnegative matrix. Through this transformation, I can reinterpret the underlying network as one where all interaction effects are positive, such that all agents conform to one another. This reinterpretation preserves the stability of equilibria, as determined by the spectral properties of $\mathbf{D}$. Leveraging this relationship, I can extend the stability properties of models with uniformly positive interactions to models that have negative interactions, yielding the following result:

PROPERTY 2: *Suppose Assumption A holds. Then, for almost all distributions $\{F_{\varepsilon|k}\}_k$, multiple locally stable equilibria exist if and only if the model has an unstable equilibrium.*

Property 2 is instrumental in the analysis. It shows that, under Assumption A, a locally stable equilibrium generically exists, and multiple locally stable equilibria exist if interaction effects are strong enough to produce an unstable equilibrium. Assumption A holds in many economically relevant settings. I provide two examples below, with more in Appendix A.1.

Example 1 (No Negative Feedback). Suppose all feedback effects are nonnegative, such that $J_{kj_1}J_{j_1j_2}\cdots J_{j_Mk} \geq 0$ for every feedback path.¹⁰ This condition holds in many familiar network structures. In a tree, it holds when all reciprocal interactions share the same sign: $J_{k\ell}, J_{\ell k} \geq 0$ or $J_{k\ell}, J_{\ell k} \leq 0$. In a circle, it holds when the network contains an even number of negative links. In a bipartite network, it holds when all interaction effects are either nonnegative or nonpositive. In a complete network, it holds when $\text{sgn}(J_{km}J_{m\ell}) = \text{sgn}(J_{k\ell})$ for every k, ℓ , and m , so that complementarity and substitutability are transitive across types.

Example 2 (Dominant Own-type Conformity). Suppose $\mathbf{D}$ is symmetric and *diagonally dominant*, so $D_{kk} \geq \sum_{\ell \neq k} |D_{k\ell}|$ for any k . In this setting, each type's response to its own average behavior is at least as strong as its combined absolute responses to the other types.

These examples show that Assumption A allows for negative interactions across a variety of network structures. It even permits negative feedback if the pressure to conform to one's own type outweighs the influence of other types, as in Example 2. The key restriction is that the accumulation of feedback effects does not lead agents to be repelled by their own type.¹¹

¹⁰This condition represents a special case of Assumption A in which the change-of-basis matrix $\mathbf{B}$ is diagonal. Specifically, for irreducible $\mathbf{J}$, it holds if and only if $\mathbf{B}\mathbf{D}(\cdot)\mathbf{B}^{-1}$ is nonnegative for some invertible, diagonal $\mathbf{B}$.

¹¹Specifically, as shown in Appendix Lemma A.6, Assumption A requires the responsiveness-weighted sum of feedback effects $J_{kj_1}J_{j_1j_2}\cdots J_{j_Mk}$ to be nonnegative, although the individual feedback effects can be negative.

2.5. Comparisons of Social Welfare across Types

I conclude this section by characterizing social welfare across different equilibria. Note that, in this model, there is no strict Pareto ranking over equilibria because extreme realizations of the idiosyncratic taste ε_i can dominate an agent's utility. Therefore, to measure welfare, I analyze the expected utility over ε_i for each agent type k evaluated at any equilibrium m .

$$\mathcal{W}_k(m) = \mathbb{E} \left(\max_{\omega_i} \left\{ h_k \omega_i + \eta_k + \sum_{\ell=1}^K J_{k\ell} \omega_i m^\ell + \varepsilon_i \omega_i + \xi_i \right\} \right). \quad (7)$$

To facilitate welfare comparisons, I rescale choices so that $\omega_i \in \{-1, 1\}$. This change does not impact behavioral outcomes; yet, it implies that, without private incentives ($h_k = \varepsilon_i = 0$ for all i, k), individual utility is the same whether everyone selects the high action or everyone selects the low action. Therefore, this rescaling ensures there is no negative externality inherent to an equilibrium when all agents are fully ambivalent between the two choices.¹²

The next property characterizes how social welfare outcomes relate to specific patterns of individual behavior. It shows that, for each type, social welfare is highest at the equilibrium in which agents are most unified in their choices. Also, if agents are privately biased toward one specific action, social welfare tends to be highest when most agents choose that action.

PROPERTY 3: *Let $\mathcal{M}$ denote the set of equilibria, and define $m^* = \operatorname{argmax}_{m \in \mathcal{M}} \mathcal{W}_k(m)$. Then $m^* = \operatorname{argmax}_{m \in \mathcal{M}} |\mathbb{E}(\bar{\omega}^k)|$. Moreover, there always exists a threshold $T_k \in \mathbb{R}$ where:*

$$m^* = \begin{cases} \operatorname{argmax}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^k), & \text{if } h_k > T_k \\ \operatorname{argmin}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^k), & \text{if } h_k < T_k. \end{cases}$$

In the presence of multiple equilibria, a key consideration for policy is how social welfare rankings differ for different types of agents. Namely, is there a single favorable equilibrium that maximizes welfare for all types, or are there tradeoffs such that improving welfare for one type would always reduce it for another? This question matters for whether policy can resolve coordination failures by shifting an economy toward a more favorable equilibrium.

In the classical model where all the interaction effects are positive, the set of equilibria is well-ordered, forming a complete lattice. From a welfare perspective, this property implies that there is always one equilibrium that maximizes social welfare for all types, provided

¹²Welfare analysis in network-based models is especially sensitive to how utility is specified, even if different specifications yield the same choices. The reason is that agents' payoffs depend on underlying drivers of spillover effects, such as conformity versus social learning, not just the effects themselves; see [Bhattacharya et al. \(2024\)](#).

that all types are privately biased toward the same action.¹³ As the next result demonstrates, this relationship does not generally apply to environments with negative interaction effects.

PROPERTY 4: *Suppose $\mathbf{J}$ is irreducible with no negative feedback. For any types k and ℓ :*

- (i) *If $J_{k\ell} > 0$, then $E(\bar{\omega}^k)$ and $E(\bar{\omega}^\ell)$ are maximized (minimized) at the same equilibrium.*
- (ii) *If $J_{k\ell} < 0$, then $E(\bar{\omega}^k)$ is maximized at the same equilibrium where $E(\bar{\omega}^\ell)$ is minimized.*

Property 4 reveals how negative interactions can introduce tradeoffs in improving welfare. For example, consider two types of agents, k and ℓ , each deciding whether to take a medical treatment, such as a vaccine. Suppose both types, in isolation, would benefit from treatment ($h_k, h_\ell > 0$), but each one prefers to differentiate from the other type ($J_{k\ell}, J_{\ell k} < 0$). If these interaction effects are strong enough, multiple equilibria exist: in one, a majority of type k is treated; in another, a majority of type ℓ is treated. Each type prefers the equilibrium where a majority of its members are treated, but both cannot achieve this outcome simultaneously.

In this setting, neither equilibrium outcome is broadly favorable. Hence, it is not obvious when a policy intervention can achieve net welfare gain. Any attempt to increase treatment uptake among one type imposes a negative externality by reducing uptake among the other.

3. IDENTIFICATION AND ESTIMATION OF ENDOGENOUS INTERACTION EFFECTS

I now show how to recover features of the interaction matrix $\mathbf{J}$ using individual choice data. The analysis proceeds in two steps. First, I prove identification when the expected choices $E(\omega_i|k)$ are observed. This allows me to focus on separating endogenous interactions from unobserved network effects. Second, I extend the analysis to the case where $E(\omega_i|k)$ is unobserved, but can be approximated by the average choice $\bar{\omega}^k$ in a finite network. I show how to construct internal instruments to correct for the resulting measurement error bias. I then derive a linear IV estimator for the interaction effects, which performs well in simulations.

3.1. Econometric Specification

Consider a setting with many networks $n = 1, \dots, N$, where agents interact only with others in their network. In practice, these networks could be schools, workplaces, neighborhoods, or villages. Suppose that the researcher has data on individual choices ω_i , either for the entire population or for a random sample of agents drawn from each network. In addition, the researcher knows the network n to which each agent belongs, as well as each agent's type k .

¹³Specifically, if $h_k > 0$ for all k , then $\arg\max_{m \in \mathcal{M}} \mathcal{W}_k(m) = \arg\max_{m \in \mathcal{M}} \mathcal{W}_\ell(m)$ for any types k and ℓ .

I allow agents' private utilities to vary systematically across networks, replacing the deterministic private payoff h_k with $h_k + \alpha_n$. Hence, for any agent i of type k in network n , private utility depends on individual-specific factors ε_i , type-specific factors h_k , and contextual network effects α_n , all of which are unobserved by the researcher. Agent i 's action is:

$$\omega_i = \mathbb{1} \left\{ h_k + \alpha_n + J_{kk} E_i(\bar{\omega}_{n,-i}^k) + \sum_{\ell \neq k} J_{k\ell} E_i(\bar{\omega}_n^\ell) + \varepsilon_i \geq 0 \right\}, \quad (8)$$

where h_k , α_n , ε_i , and $\{J_{k\ell}\}_{k,\ell}$ are unknown parameters. The model may also be extended to include observed covariates or prices at the agent- or network-level; however, as these do not meaningfully impact the results, I omit them for now and give details in Appendix B.1.

I maintain two assumptions about the error structure. First, I assume the random payoffs ε_i and ε_j are independent for any agents i and j within and across networks. Hence, there is no covariation in the error terms. Second, I assume that, given an agent's type k , the error term ε_i is independent of α_n . This condition rules out self-selection into networks based on unobserved idiosyncratic payoffs.¹⁴ Both of these assumptions are standard in the literature on social interactions, and they can also be relaxed to allow for selection on observables.¹⁵

ASSUMPTION B.1: (i) $\{\varepsilon_i\}_i$ are mutually independent; (ii) $P(\varepsilon_i \leq z | k, \alpha_n) = F_{\varepsilon|k}(z)$.

As before, all agents act with incomplete information and consistent beliefs. In equilibrium:

$$m_n^k = F_{\varepsilon|k} \left(h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell} m_n^\ell \right), \quad \text{for } k = 1, \dots, K \text{ and } n = 1, \dots, N. \quad (9)$$

As shown in Section 2, sufficiently strong interaction effects can generate multiple equilibria. In such cases, I assume that agents correctly anticipate which equilibrium is realized in their network.¹⁶ Equilibrium multiplicity does not, by itself, interfere with identification: a given parameter vector may admit multiple equilibria even while the observed choice distribution uniquely determines the parameters of interest. My identification analysis uses the restrictions that the equilibrium conditions (9) impose, without ever directly solving for eq-

¹⁴This restriction is stronger than needed for identification. As shown by Manski (1988) and also discussed by Horowitz (2009), full conditional independence can be replaced by a weaker quantile independence assumption.

¹⁵Let $P(\varepsilon_i \leq z | W_n, \alpha_n, k) = F_{\varepsilon|W_n,k}(z)$, where W_n is an observable network characteristic. Even if $F_{\varepsilon|W_n,k}$ does not equal $F_{\varepsilon|k}$, the identification arguments presented still follow by comparing networks with the same W_n .

¹⁶This assumption can be rationalized by noting that, whenever the realized equilibrium is locally stable, agents can compute it independently through fixed-point iteration on (9), as long as they begin with a common prior m_0 .

equilibrium outcomes. In doing so, I demonstrate that the parameters of interest are identified and may be consistently estimated regardless of which equilibrium is realized in the data.¹⁷

In most cases, it is natural to assume that the equilibrium is locally stable (see Section 2.4 for conditions); otherwise, it is unlikely to be realized in practice. Yet, although stability can matter for interpretation, the identification analysis does not explicitly rely on this property.

3.2. Identification with Known Expected Average Choices

Suppose that researchers observe the expected average choices m_n^k . Then, the main barrier to identification is the unobserved network effect α_n , which obscures the effect of social interactions on group-level outcomes. This issue is noted by [Brock and Durlauf \(2007\)](#), Prop. 2, who show that network unobservables prevent point identification of interaction effects.

I propose a partial identification strategy to recover key features of $\mathbf{J}$ without placing any added assumptions on α_n . This method involves differencing-out α_n among different types of agents in the same network. The intuition behind this strategy is captured in the following lemma. I give the proof alongside the result so the approach can be more clearly interpreted.

LEMMA 2—Sufficiency Claim: $E(\omega_i|k, \alpha_n, \{m_n^\ell\}_\ell) = E(\omega_i|k, \{m_n^\ell\}_\ell)$ for all n and k .

Proof. Fix any $\tilde{k} \neq k$. Since $F_{\varepsilon|\tilde{k}}$ is strictly increasing, equation (9) can be inverted so that:

$$\alpha_n = F_{\varepsilon|\tilde{k}}^{-1}(m_n^{\tilde{k}}) - h_{\tilde{k}} - \sum_{\ell=1}^K J_{\tilde{k}\ell} m_n^\ell. \quad (10)$$

Substituting this expression into the definition of $E(\omega_i|k, \alpha_n, \{m_n^\ell\}_\ell)$ yields the equation:

$$E(\omega_i|k, \alpha_n, \{m_n^\ell\}_\ell) = F_{\varepsilon|k} \left(h_k - h_{\tilde{k}} + \sum_{\ell=1}^K (J_{k\ell} - J_{\tilde{k}\ell}) m_n^\ell + F_{\varepsilon|\tilde{k}}^{-1}(m_n^{\tilde{k}}) \right). \quad (11)$$

The network effect cancels. Thus, $E(\omega_i|k, \alpha_n, \{m_n^\ell\}_\ell)$ is a deterministic function of $\{m_n^\ell\}_\ell$.
Q.E.D.

This lemma shows how observable differences among agents in the same network may be used to control for contextual effects. Consider two agents i and j of different types, k_1 and k_2 , who both belong to network n . Because they reside in the same network, all contextual factors affecting i also apply to j . Any difference in i and j 's choices is due to differences in type-specific preferences, $(h_{k_1}, \{J_{k_1\ell}\}_\ell)$ and $(h_{k_2}, \{J_{k_2\ell}\}_\ell)$, or idiosyncratic tastes, ε_i and ε_j .

¹⁷This contrasts with the two-step approaches used in the empirical industrial organization literature, where identification relies on computing equilibrium conditions. See [Bhattacharya et al. \(2024\)](#), Section 5.5, for discussion.

This framework offers a natural panel structure, allowing me to control for network-level factors by comparing the outcomes of different agents in the same network. This approach is conceptually similar to the method of differencing-out incidental parameters in panel data models; see, for example, [Chamberlain \(1980\)](#).¹⁸ By removing α_n , I achieve identification of the differences $h_{k_1} - h_{k_2}$ and $\{J_{k_1\ell} - J_{k_2\ell}\}_\ell$, which capture how agents differ in their private incentives and responses to others' behavior. At the same time, this approach leaves the levels of h_k and $J_{k\ell}$, as well as the network effects themselves, unidentified, which, as [Chamberlain \(1984\)](#) discusses, may limit the range of probability statements one can make.

I present two versions of the identification result. First, I establish semiparametric identification when the error distributions $F_{\varepsilon|k}$ are known. Second, I establish nonparametric identification when $F_{\varepsilon|k}$ are unknown. While the latter result is more flexible, it requires more variation in the data. In both versions, identification is achieved without α_n being observed.

Conditions for Semiparametric Identification

Before stating the semiparametric identification result, I present the following assumption.

ASSUMPTION B.2: There is sufficient variation across networks for $\text{rank}(\text{Var}(m_n)) = K$.

This rank condition guarantees that there is enough variation in α_n within the population to uncover the nonlinear relationship between m_n and α_n . This nonlinearity is a core feature of discrete choice models, and it marks a key distinction from linear simultaneous equations models, where identification is typically impeded by the *reflection problem* ([Manski, 1993](#)). In linear models, m_n and α_n are linearly dependent. Thus, even if α_n is fully determined by observables, one cannot isolate social interactions without additional exclusion restrictions.

By contrast, the reflection problem does not arise in discrete choice settings. As shown by [Brock and Durlauf \(2001, 2007\)](#), discrete choices generate a nonlinear relationship between m_n and α_n , which acts as a built-in exclusion restriction. This nonlinearity allows me to isolate endogenous interactions from contextual network effects. Neither continuous nor unbounded support of α_n is necessary. It suffices that the covariance matrix of m_n has full rank.

THEOREM 1: Suppose Assumptions B.1 and B.2 hold, and assume m_n^k is observed for all networks n and agent types k . If the distribution functions $\{F_{\varepsilon|k}\}_k$ are identified, then:

- (i) Without further assumptions, $\{h_{k_1} - h_{k_2}\}_{k_1, k_2}$ and $\{J_{k_1\ell} - J_{k_2\ell}\}_{k_1, k_2, \ell}$ are identified.
- (ii) If $\alpha_n = W_n' d$ for some observed vector W_n , then d , $\{h_k\}_k$, and $\{J_{k\ell}\}_{k, \ell}$ are identified.

¹⁸[Bonhomme \(2012\)](#) proposes a systematic approach to obtain moment restrictions that are free of fixed effects via functional differencing. Additionally, see [Bonhomme and Dano \(2023\)](#) and [Dano \(2025\)](#) for recent advances.

This theorem has two parts. Part (i) draws on Lemma 2 to show that the model is partially identified even if α_n is unobserved. In this case, the differences $h_{k_1} - h_{k_2}$ and $\{J_{k_1\ell} - J_{k_2\ell}\}_\ell$ are identified for all (k_1, k_2) .¹⁹ This result is new to the literature and is the theorem's main contribution. Part (ii) covers a special case, studied by Brock and Durlauf (2001) and others, where α_n is a fixed, linear function of observables. In that case, the model is point identified.

Conditions for Nonparametric Identification

For nonparametric identification, I require an extra assumption: there must be an exogenous individual-level covariate with continuous, unbounded support. Variation in this covariate traces out the error distributions $F_{\varepsilon|k}$, which I then use to recover the remaining parameters.

To set ideas, I first modify the choice equation to include exogenous covariates, $X_i \in \mathbb{R}^r$. Let $\omega_i = \mathbb{1}\{X_i'c + h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell}m_n^\ell + \varepsilon_i \geq 0\}$, with $P(\varepsilon_i \leq z|X_i, k, \alpha_n) = F_{\varepsilon|k}(z)$ and $P(X_i \leq x|k, \alpha_n) = P(X_i \leq x|k)$. Suppose that X_i satisfies the following assumption:

ASSUMPTION B.3: For every k , $\text{supp}(X|k)$ is not contained in a proper affine subspace of $\mathbb{R}^r$; also, there is a component x_j of X —with a nonzero coefficient c_j —such that, for almost all $x_{-j|k}$, the distribution of $x_{j|k}$ given $x_{-j|k}$ has positive density everywhere on $\mathbb{R}$.

This assumption, adapted from Manski (1988), ensures that there is sufficient variation in at least one individual-level covariate to recover the error distributions $F_{\varepsilon|k}$ up-to-scale, even when the functional forms are unknown. With this assumption, I obtain the following result:

THEOREM 2: Suppose Assumptions B.1, B.2, and B.3 hold, and assume m_n^k is observed for all networks n and agent types k . Then $\{F_{\varepsilon|k}\}_k$ and c are identified up-to-scale. Also:

- (i) Assuming nothing else, $(\{h_{k_1} - h_{k_2}\}_{k_1, k_2}, \{J_{k_1\ell} - J_{k_2\ell}\}_{k_1, k_2, \ell})$ is identified up-to-scale.
- (ii) If $\alpha_n = W_n'd$ for some observed W_n , then $(d, \{h_k\}_k, \{J_{k\ell}\}_{k, \ell})$ is identified up-to-scale.

3.3. Identification with Unknown Expected Average Choices

In the data, the researcher observes individual choices, but not the equilibrium expectations m_n^k that enter agents' decisions. In particular, agents respond to the expected average behavior of each type, which need not equal the average choice $\bar{\omega}_n^k$ realized in a finite network.

This distinction between expected and realized behavior rests on substantive assumptions

¹⁹If the error distributions are known, then $J_{k\ell}$ is identified up-to-location by normalizing the own-type effects J_{kk} to one. Similarly, private payoffs h_k are identified up-to-location by setting $h_{k^*} = 0$ for a reference type k^* .

about the information structure. As discussed in Section 2.3, agents know their own private idiosyncratic tastes, but do not observe those of their peers. When making decisions, they respond to their expectations of each type's average behavior, with beliefs that are assumed to be consistent in equilibrium. This framework is motivated by economic settings in which agents make choices with private information, without being able to perfectly anticipate everyone else's contemporaneous actions. It is standard in the literature following [Brock and Durlauf \(2001\)](#), and it simplifies the identification analysis by yielding an equilibrium characterized in terms of choice probabilities.²⁰ Nevertheless, this assumption need not have a principled basis in a given setting; I discuss its suitability for each application in Section 4.

For researchers, the assumed information structure creates a measurement problem: agents respond to expectations that can only be measured imperfectly from observed choices. In particular, with finite networks, the model implies that $\bar{\omega}_n^k = m_n^k + u_n^k$, where u_n^k denotes measurement error. If the researcher substitutes $\bar{\omega}_n^k$ for m_n^k without accounting for this error, the preceding identification arguments no longer directly apply. To see why, rewrite (8) as:

$$\omega_i = \mathbb{1} \left\{ h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell} \bar{\omega}_n^\ell + \tilde{\varepsilon}_i \geq 0 \right\}, \quad \text{where} \quad \tilde{\varepsilon}_i = \varepsilon_i - \sum_{\ell=1}^K J_{k\ell} u_n^\ell. \quad (12)$$

In this equation, the observed average choices $\{\bar{\omega}_n^\ell\}_\ell$ are endogenous because they are correlated with the error term $\tilde{\varepsilon}_i$. Even if agent i 's choice ω_i were excluded when constructing $\bar{\omega}_n^k$, it would still be that $\text{Cov}(\bar{\omega}_n^\ell, \tilde{\varepsilon}_i) = -J_{k\ell} \times \text{Var}(u_n^\ell) \neq 0$ for any type ℓ . Consequently, treating realized averages as equilibrium expectations can bias estimates of interaction effects.

The remainder of this subsection shows how to correct for measurement error under the incomplete information structure. In particular, I show how to recover the parameters of interest from choice data when networks are finite and agents' expectations are unobserved.

Constructing Instruments to Correct for Measurement Error

I propose a way to correct for measurement error using internal instruments. This approach involves two steps. First, in each network n and type k , I randomly partition agents into two subsamples, a and b . Second, I compute the average actions, $\bar{\omega}_{n,a}^k$ and $\bar{\omega}_{n,b}^k$, in each subsample and estimate the model via IV, treating $\bar{\omega}_{n,a}^k$ as the regressor and $\bar{\omega}_{n,b}^k$ as the instrument.

By construction, $\bar{\omega}_{n,a}^k$ and $\bar{\omega}_{n,b}^k$ are noisy proxies for m_n^k with sampling errors that are independent. This independence follows from the assumed information structure, under which agents respond to expected rather than realized choices. Therefore, either proxy can instru-

²⁰See [de Paula \(2013\)](#) for a careful review of how information assumptions affect identification of these models.

ment for the other, yielding consistent estimates as the number of networks grows, for fixed network sizes. Before stating the identification result, I will first motivate it with an example.

Example (Brock-Durlauf Model). Consider the model studied by [Brock and Durlauf \(2001\)](#):

$$m_n = \tanh(h + W_n' d + J m_n), \quad \text{for } n = 1, \dots, N, \quad (13)$$

where W_n is an observed vector and h , d , and J are unknown parameters.²¹ When taking the model to data, it is common practice to replace equilibrium expectations m_n with observed averages $\bar{\omega}_n$, thus treating a noisy finite-network proxy as if it were observed without error.

Without accounting for measurement error, standard estimators, such as maximum likelihood or OLS, deliver biased estimates. To see how, consider an OLS estimand based on the observed means $\bar{\omega}_n$. Defining $u_n = \bar{\omega}_n - m_n$ and $v_n = \tanh^{-1}(\bar{\omega}_n) - \tanh^{-1}(m_n)$, I write:

$$\tanh^{-1}(\bar{\omega}_n) = h + W_n' d + J \bar{\omega}_n + \tilde{\xi}_n, \quad \text{where } \tilde{\xi}_n = -J u_n + v_n. \quad (14)$$

To ease notation, let $\tilde{m}_n$, $\tilde{\omega}_n$, and $\tilde{Y}_n$ denote the residuals from a least squares regression of m_n , $\bar{\omega}_n$, and $\tanh^{-1}(\bar{\omega}_n)$, respectively, on the vector $(1, W_n')$. The OLS estimand for J is:

$$J^{\text{OLS}} = \frac{\text{Cov}(\tilde{\omega}_n, \tilde{Y}_n)}{\text{Var}(\tilde{\omega}_n)} = J \times \frac{\text{Var}(\tilde{m}_n)}{\text{Var}(\tilde{m}_n) + \text{Var}(u_n)} + \frac{\text{Cov}(\tilde{\omega}_n, v_n)}{\text{Var}(\tilde{m}_n) + \text{Var}(u_n)}. \quad (15)$$

This estimand suffers from two sources of bias, since both the regressor and the outcome are measured with error. In finite networks, OLS therefore fails to recover the interaction effect.

As illustrated in Figure 1, the bias can be substantial even for very large networks. When setting $J = 0.5$, OLS overstates the interaction effect by a factor of five in networks with 100 agents and by a factor of two in networks with 500 agents. The bias only becomes small when network sizes reach 5,000 agents—a scale rarely attainable in empirical applications.

To overcome this issue, I propose IV estimation: splitting each network into two parts, a and b , and using the average choice in part b as an instrument for the average choice in part a . By construction, the instrument $\bar{\omega}_{n,b}$ is uncorrelated with measurement error in $\bar{\omega}_{n,a}$, and instrument relevance holds, as $\text{Cov}(\bar{\omega}_{n,a}, \bar{\omega}_{n,b}) = \text{Var}(m_n) \neq 0$. By standard arguments, J is identified by: (1) regressing $\bar{\omega}_{n,a}$ on $(1, W_n', \bar{\omega}_{n,b})$ in the first stage to obtain fitted values $\hat{\bar{\omega}}_{n,a}$ and (2) regressing the outcome Y_n on $(1, W_n', \hat{\bar{\omega}}_{n,a})$ in the second stage. To guarantee that the estimator jointly identifies J , d , and h , I address a further complication: owing to the nonlinearity of $\tanh^{-1}$, the measurement error in $\tanh^{-1}(\bar{\omega}_n)$ is nonzero in expectation.

²¹The authors allow for individual-level covariates X_i , which I exclude in this example for notational simplicity.

This issue can be corrected by defining Y_n as a linear combination of $\tanh^{-1}(\bar{\omega}_{n,a})$ across subsamples, exploiting the known dependence between $E(v_{n,a})$ and the subsample size. As demonstrated below, the procedure jointly recovers J , d , and h even with finite networks.

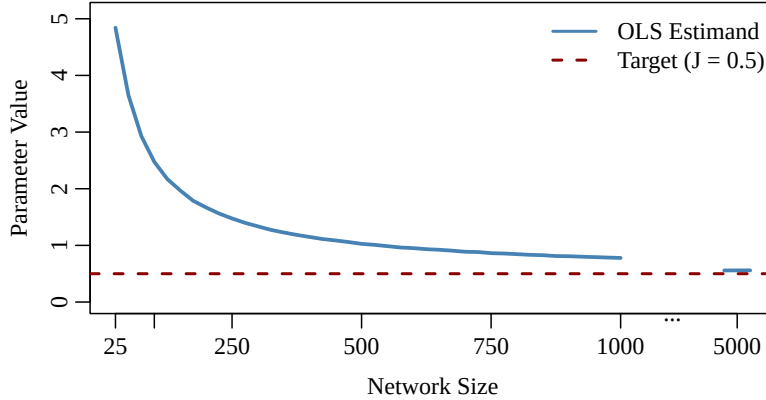


FIGURE 1.—OLS estimand for J across network sizes ($J = 0.5$, $h = 1$, $d = 1$); same design as in Table 2(A).

An IV Estimand for Endogenous Interaction Effects

To formalize the identification result, first define $\bar{\omega}_{n,a}^k$ and $\bar{\omega}_{n,b}^k$ to be the average choices in each randomly selected subset of network n and type k . Throughout the analysis, I define $\bar{\omega}_{n,a}^k$ as the endogenous regressor and $\bar{\omega}_{n,b}^k$ as the excluded instrument. Define the outcomes:

$$Y_n^k = \Omega_n^k F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k) + (1 - \Omega_n^k) F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a'}^k), \quad (16)$$

where a' is a strict subset of a , and Ω_n^k is a known weight that depends on the sizes of a and a' , denoted by $I_{n,a}^k$ and $I_{n,a'}^k$, respectively. To define this weight, I consider a sequence of partitioning rules p , where $\kappa(p, n) \in \{a, a'\}$ indicates which subset the rule selects in network n . For the first P rules, define the matrix $\mathbf{E}_{k,P}$ by $[\mathbf{E}_{k,P}]_{i,j} = E[(I_{n,\kappa(p,n)}^k)^{1-i}]$, and set $\Omega_n^k = \lim_{P \rightarrow \infty} \sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} \mathbb{1}\{\kappa(p, n) = a\}$. The Appendix gives details on rules and sampling conditions under which the matrices are nonsingular and the resulting outcomes yield valid IV moments. While the formal identification argument takes the limit as $P \rightarrow \infty$, I find that in practice, even $P = 1$ or $P = 2$ suffices to deliver estimates with effectively zero bias.

THEOREM 3: *Suppose Assumptions B.1 and B.2 hold and $\{F_{\varepsilon|k}\}_k$ are identified. Then:*

- (i) $\beta_{k_1, k_2} = [h_{k_1} - h_{k_2}, J_{k_1 1} - J_{k_2 1}, \dots, J_{k_1 K} - J_{k_2 K}]'$ is identified for any (k_1, k_2) . Also:

$$\beta_{k_1, k_2} = E(Z_n X_n')^{-1} E(Z_n [Y_n^{k_1} - Y_n^{k_2}]), \quad \text{where } \begin{cases} X_n = (1, \bar{\omega}_{n,a}^1, \dots, \bar{\omega}_{n,a}^K)' \\ Z_n = (1, \bar{\omega}_{n,b}^1, \dots, \bar{\omega}_{n,b}^K)' \end{cases} \quad (17)$$

(ii) If $\alpha_n = W_n' d$ for observed W_n , $\beta_k = [h_k, d', J_{k1}, \dots, J_{kK}]'$ is identified for any k . Also:

$$\beta_k = E(Z_n X_n')^{-1} E(Z_n Y_n^k), \quad \text{where } \begin{cases} X_n = (1, W_n', \bar{\omega}_{n,a}^1, \dots, \bar{\omega}_{n,a}^K)' \\ Z_n = (1, W_n', \bar{\omega}_{n,b}^1, \dots, \bar{\omega}_{n,b}^K)' \end{cases} \quad (18)$$

Theorem 3 shows that the endogenous interaction effects are identified from data on finite networks. These effects can be estimated by linear IV, where $\bar{\omega}_{n,b}^k$ is the instrument for $\bar{\omega}_{n,a}^k$, and outcomes are defined in (16). Replacing population moments with sample counterparts yields estimators that are consistent as the number of networks grows, for a fixed network size distribution.²² Appendix B.3 derives their asymptotic properties to support inference.

In Table 2, I assess the performance of the IV estimator using Monte Carlo simulations, varying both the network sizes and the number of networks. The results highlight two key properties of the estimator. First, performance improves for larger network sizes. This is because, with larger networks, $\bar{\omega}_{n,a}^k$ more closely approximates the expectation m_n^k , reducing measurement error. Second, holding network sizes fixed, performance improves as the number of networks increases. This pattern is consistent with the identification arguments underlying Theorem 3, which shows that IV can recover interaction effects from data on finite networks. Crucially, this same pattern does not hold for OLS. In particular, while OLS remains biased as the number of networks tends to infinity, the IV estimator yields increasingly precise estimates. Moreover, the IV estimator's confidence intervals achieve coverage rates tightly centered around the nominal 95% level, whereas OLS coverage is nearly zero.

While small network sizes do not preclude identification, one practical consideration in finite samples is instrument strength. For small network-type cells, the split-sample means, $\bar{\omega}_{n,a}^k$ and $\bar{\omega}_{n,b}^k$, become noisier proxies for the common expectation m_n^k , weakening the population first-stage relationship between them. Thus, reliable estimation may require more networks to estimate this weaker first-stage relationship with sufficient precision when the number of agents per network and type is small. Appendix Table A.1 illustrates this tradeoff by extending the simulation design in Table 2 to very small networks with 4 to 10 agents per type, showing that the estimator can perform well if there are sufficiently many networks.

4. EMPIRICAL APPLICATIONS

I now present two empirical applications of the model: peer effects in Tennessee primary schools and spillovers in deworming treatment uptake in Kenya. In both settings, allowing for nonuniform interactions materially changes the conclusions about the role of peer influence.

²²Consistency requires that the measurement errors are mean zero and uncorrelated, such that $E(u_{n,a}^k) = 0$ and $E(u_{n,a}^k u_{n,b}^\ell) = 0$ for all k, ℓ . These moment conditions continue to hold even in the presence of multiple equilibria.

TABLE 2
PERFORMANCE OF OLS AND IV ESTIMATORS BY NETWORK SIZE AND NUMBER OF NETWORKS

Number of Networks	Agents per Network	Specification A ($K = 1$)						Specification B ($K = 2$)					
		$\hat{h}$		$\hat{d}$		$\hat{J}$		$\widehat{h_1 - h_2}$		$\widehat{J_{11} - J_{21}}$		$\widehat{J_{22} - J_{12}}$	
		OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV
N = 250	30	8.62	4.72	0.43	0.27	16.66	8.70	1.60	1.27	7.61	4.45	3.12	1.54
		(0.20)	(2.50)	(0.05)	(0.59)	(0.27)	(3.40)	(0.17)	(1.16)	(0.28)	(2.16)	(0.17)	(1.23)
		[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.98]	[0.00]	[0.98]	[0.00]	[0.97]
	100	2.00	0.76	0.09	0.05	3.91	1.36	0.36	0.16	1.39	0.56	0.37	0.24
		(0.22)	(0.95)	(0.06)	(0.22)	(0.30)	(1.29)	(0.14)	(0.49)	(0.25)	(0.89)	(0.15)	(0.51)
		[0.00]	[0.98]	[0.00]	[0.95]	[0.00]	[0.98]	[0.00]	[0.99]	[0.00]	[0.98]	[0.04]	[0.96]
	200	0.72	0.25	0.04	0.02	1.39	0.43	0.11	0.06	0.41	0.20	0.11	0.09
		(0.17)	(0.51)	(0.04)	(0.12)	(0.23)	(0.70)	(0.10)	(0.28)	(0.18)	(0.51)	(0.11)	(0.30)
		[0.00]	[0.96]	[0.03]	[0.93]	[0.00]	[0.97]	[0.05]	[0.97]	[0.03]	[0.98]	[0.18]	[0.95]
N = 1,000	30	8.62	1.04	0.44	0.06	16.63	1.91	1.61	0.28	7.70	0.99	3.14	0.33
		(0.10)	(1.10)	(0.03)	(0.26)	(0.14)	(1.49)	(0.08)	(0.55)	(0.14)	(1.02)	(0.09)	(0.58)
		[0.00]	[0.98]	[0.00]	[0.97]	[0.00]	[0.98]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.96]
	100	2.00	0.22	0.09	0.01	3.91	0.39	0.37	0.05	1.43	0.15	0.37	0.06
		(0.11)	(0.47)	(0.03)	(0.11)	(0.15)	(0.64)	(0.07)	(0.24)	(0.12)	(0.45)	(0.07)	(0.26)
		[0.00]	[0.95]	[0.00]	[0.93]	[0.00]	[0.95]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.96]
	200	0.72	0.07	0.03	0.00	1.38	0.12	0.11	0.01	0.43	0.04	0.11	0.02
		(0.09)	(0.26)	(0.02)	(0.06)	(0.12)	(0.35)	(0.05)	(0.14)	(0.09)	(0.25)	(0.05)	(0.15)
		[0.00]	[0.95]	[0.00]	[0.93]	[0.00]	[0.95]	[0.00]	[0.98]	[0.00]	[0.98]	[0.00]	[0.96]
N = 5,000	30	8.64	0.19	0.44	0.01	16.67	0.34	1.61	0.07	7.71	0.24	3.15	0.08
		(0.04)	(0.48)	(0.01)	(0.11)	(0.06)	(0.65)	(0.04)	(0.25)	(0.06)	(0.46)	(0.04)	(0.26)
		[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.94]	[0.00]	[0.94]	[0.00]	[0.94]
	100	2.02	0.05	0.09	0.00	3.94	0.09	0.37	0.01	1.44	0.03	0.37	0.02
		(0.05)	(0.21)	(0.01)	(0.05)	(0.07)	(0.29)	(0.03)	(0.11)	(0.06)	(0.20)	(0.03)	(0.11)
		[0.00]	[0.94]	[0.00]	[0.92]	[0.00]	[0.94]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.94]
	200	0.71	0.01	0.03	0.00	1.38	0.02	0.11	0.00	0.43	0.01	0.11	0.00
		(0.04)	(0.12)	(0.01)	(0.03)	(0.05)	(0.16)	(0.02)	(0.06)	(0.04)	(0.11)	(0.02)	(0.07)
		[0.00]	[0.96]	[0.00]	[0.93]	[0.00]	[0.96]	[0.00]	[0.97]	[0.00]	[0.97]	[0.00]	[0.95]

Notes. This table reports mean squared errors, average estimated standard errors (in parentheses), and 95% confidence interval coverage rates (in brackets), based on 1,000 simulation draws. Specification A estimates β_k , defined in equation (18), with $K = 1$ and $\alpha_n = W_n' d$, using the same design as Figure 1. Specification B estimates β_{k_1, k_2} , defined in equation (17), with $K = 2$ and α_n treated as unobserved. X_n and Z_n are constructed from disjoint 2/3 and 1/3 subsamples, respectively, with $P = 1$. Appendix C.1 describes the simulation design and parameter values.

4.1. Project STAR: Gender Differences in Peer Effects

The first application studies the Project STAR class size reduction experiment, launched in 1985 among kindergarteners at 79 public schools in Tennessee. In each school, students and teachers were randomly assigned to one of three classroom types: small (13–17 students), large (22–25), and large with a teacher's aide. At the end of the year, students took exams in math and reading. My analysis uses data from the first-year cohort, which covered 6,325 students across 325 classrooms. See Word et al. (1990) for a detailed summary of the program.

To learn about gender differences in peer effects, I study a model with two student types: male (m) and female (f). The Project STAR sample is well-suited to this decomposition, as boys and girls are both well-represented in all classrooms.²³ As outcomes, I use indicators for whether a student scored in the top 50% statewide in math and in reading.²⁴ Because scores are realized only at the end-of-year exam, students can respond only to their classmates' expected performance, consistent with the model's incomplete information assumption.

Outcomes are given by $\omega_i = \mathbb{1}\{e_i \geq \varepsilon_i\}$, where e_i is a student's effort and ε_i is a threshold that depends on individual ability, with distribution $F_{\varepsilon|k}$. Students choose effort to maximize:

$$U_i(e_i|k, n) = \left(h_k + \alpha_n + J_{kk} E_i(\bar{\omega}_{n,-i}^k) + J_{k\ell} E_i(\bar{\omega}_n^\ell) \right) e_i - \frac{1}{2} e_i^2. \quad (19)$$

The gender fixed effect h_k captures systematic differences in cognitive development and prior socialization. The classroom fixed effect α_n accounts for contextual factors such as teacher quality and class size. The gender-specific peer effects $J_{k\ell}$ measure how effort depends on the expected performance of male and female peers. Equation (9) defines an equilibrium.

Under the experimental protocols, students within a school could not self-select into classrooms. Thus, Assumption B.1 holds conditional on school. To account for nonrandom assignment into schools, I include school fixed effects in the estimation.²⁵ When implementing the estimators, I assume ε_i has a logistic distribution, $F_{\varepsilon|k}(z) = [1 + \exp(-z)]^{-1}$, thus avoiding the practical challenge of nonparametric estimation with a relatively small sample.

Table 3 shows IV estimates of $h_m - h_f$, $J_{mm} - J_{fm}$, and $J_{ff} - J_{mf}$ using Project STAR data. These estimates measure differences by gender in private effort costs and peer effects, while controlling for all factors at the classroom-level. The estimated difference between h_m and h_f is tightly centered around zero, implying that boys and girls perform similarly in the absence of peer effects. Peer effects, however, vary significantly by gender, with $J_{mm} > J_{fm}$ and $J_{ff} > J_{mf}$, indicating that students conform more strongly to peers of the same gender. These differences are slightly larger for reading than for math, though the differences across outcomes are not statistically significant. I also cannot reject that $J_{mm} - J_{fm} = J_{ff} - J_{mf}$, which suggests that peer effect differences may be approximately symmetric across gender.

To assess how allowing for generalized interactions affects the results, I investigate, in Table A.3 of the Appendix, how the estimates change under the assumption of uniform peer

²³ As the data lack classroom identifiers, I follow [Boozer and Cacciola \(2001\)](#) and [Graham \(2008\)](#) by uniquely matching students to classes using class observables. This yields a sample of 6,248 students across 321 classrooms.

²⁴ Appendix Table A.2 shows that the estimates remain similar for alternative thresholds, such as 25% and 75%.

²⁵ Following [Graham \(2008\)](#), I also report estimates for a subsample of schools that had only one classroom of each type. This addresses the unlikely possibility of nonrandom assignment to classrooms within classroom types.

effects ($J_{k\ell} = J$). This restriction delivers substantively different conclusions, such as estimating that girls have lower private effort costs than boys ($h_f > h_m$), though this difference becomes statistically insignificant after including school fixed effects. Moreover, restricting peer effects to be uniform has implications for the predicted impacts of classroom-level policies. Since $J_{mm} > J_{fm}$ and $J_{ff} > J_{mf}$, a policy that boosts the achievement of boys would have outsized effects on other boys but limited effects on girls. The overall impact of a classroom intervention can therefore depend substantially on which students it directly targets.

To further validate the approach, I conduct three robustness analyses. First, in Appendix Table A.5, I test whether the parameters ($h_f, h_m, J_{ff}, J_{fm}, J_{mf}, J_{mm}$) depend on observed classroom characteristics, such as the poverty rate, minority share, location, or teacher education and experience. Such dependence would suggest the model is misspecified. I find no evidence that these parameters depend on classroom factors. Second, in Appendix Figure A.1, I assess the sensitivity of the IV estimates to alternative classroom partitioning rules, plotting histograms of estimates across different partitions. I find that the estimates are relatively stable across partitioning rules. Third, in Appendix Table A.6, I validate the IV estimator's finite-sample performance using simulations calibrated to Project STAR's number of classrooms, classroom sizes and gender composition, and the estimated interaction effects.

TABLE 3
PROJECT STAR: IV ESTIMATES OF PEER EFFECT DIFFERENCES BY GENDER

	Top 50% Math Score		Top 50% Reading Score	
	(1)	(2)	(1)	(2)
$h_m - h_f$	-0.112 (0.16)	0.082 (0.20)	-0.029 (0.18)	0.019 (0.29)
$J_{mm} - J_{fm}$	4.229 (0.86)	4.755 (0.76)	5.339 (0.82)	4.975 (0.55)
$J_{ff} - J_{mf}$	4.127 (1.01)	4.789 (1.07)	5.223 (1.09)	4.686 (0.81)
Observations	5,798	5,798	5,718	5,718
School FE	No	Yes	No	Yes
1st Stage $F(\bar{\omega}_n^m)$	29.40	30.06	25.52	25.73
1st Stage $F(\bar{\omega}_n^f)$	19.77	19.72	13.37	12.98

Notes. This table presents estimates of gender differences in peer effects, applying the estimand in (17). Standard errors use the asymptotic variance $Q\Omega Q'$ derived in Appendix B.3, where $\Omega = \text{Var}(Z_n[Y_n^{k_1} - Y_n^{k_2} - X_n'\beta_{k_1, k_2}])$ and $Q = E(Z_n X_n')^{-1}$. X_n and Z_n are constructed using disjoint 2/3 and 1/3 sample splits, respectively, setting $P = 1$.

4.2. Primary School Deworming: Treatment Uptake with Nonuniform Spillovers

The second application revisits Miguel and Kremer's (2004) seminal study of the Primary School Deworming Project, conducted in Busia, Kenya, from 1998 to 2001. In this study, 75 schools were randomly assigned to three equal-sized groups: Group 1 began receiving free deworming treatment in 1998, Group 2 in 1999, and Group 3 not until 2001. A key feature

of the intervention was that individuals could opt out of treatment. Indeed, among the nearly 15,000 students eligible in 1998-1999, only about 75% received treatment in a given year. This partial compliance provides a natural setting to analyze spillovers in treatment uptake.

Following the original paper, I focus on the years 1998 and 1999, and I restrict the sample to school-aged children (ages 6-18) eligible for treatment. Eligibility covered all students at treatment schools, except girls older than 13 and, in 1999, anyone without parental consent.

I model each school as a social network in which interactions are global, and I allow the interactions to vary across age and gender. Given that girls older than 13 were ineligible for treatment, I specify a model with three types of students: girls aged ≤ 13 , boys aged ≤ 13 , and boys aged > 13 . Since most schools were large (averaging over 400 students), I assume students cannot track everyone else's choice and therefore act with incomplete information.

In each year t , a student i of type k in school n decides whether to accept treatment based on private benefits and social interaction effects. A student's treatment decision is given by:

$$\omega_{i,t} = \mathbb{I} \left\{ h_{k,t} + \alpha_{n,t} + X'_{i,t}c + J_{kk} E_i(\bar{\omega}_{n,-i,t}^k) + \sum_{\ell \neq k} J_{k\ell} E_i(\bar{\omega}_{n,t}^\ell) + \varepsilon_{i,t} \geq 0 \right\}. \quad (20)$$

The time-varying type fixed effect $h_{k,t}$ captures systematic differences by age and gender in health risk and other determinants of treatment uptake. The time-varying school fixed effect $\alpha_{n,t}$ accounts for any cross-school externalities, as specified by Miguel and Kremer (2004), as well as school-level differences in treatment provision, baseline infection rates, and other factors.²⁶ The model also includes student-level controls $X_{i,t}$ for prior treatment uptake and prior school attendance, as well as an idiosyncratic taste term $\varepsilon_{i,t}$, with distribution $F_{\varepsilon|k,t}$.²⁷

The interaction effects $J_{k\ell}$ measure how uptake depends on the expected treatment rates of different types of peers. Since deworming medication was both a new technology and a public good (by reducing infection rates among untreated individuals), social interactions in this setting reflect a combination of peer effects, social learning, and free-riding motives.²⁸

²⁶In the paper, cross-school spillovers operate through the number of students living near the school who attend other treatment schools. As this channel is school $\times$ time-specific, it is fully accounted for by the fixed effect $\alpha_{n,t}$.

²⁷I control for prior uptake to isolate contemporaneous interactions from preexisting factors. In Appendix Table A.7, I also report estimates restricted to first-time eligible students. I control for attendance to ensure spillovers operate through uptake, not attendance, given that absence on the treatment day was a key source of noncompliance. Note that only prior attendance is used, as current attendance might be endogenously affected by treatment uptake.

²⁸Kremer and Miguel (2007) also study peer effects in this setting; yet, their analysis differs in three respects. First, they study parents' consent decisions, while I study student uptake after parental consent is granted. Second, they examine local interactions among close social contacts in a small sample of parents, where spillovers operate via the *exogenous* number of links to treatment schools. Instead, I study *endogenous* interactions within the school, controlling for the number of nearby students attending treatment schools. Third, they focus on later years of the

Under the experimental design, students could not self-select into schools based on treatment, and most schools had similar observable characteristics (Miguel and Kremer, 2004). Nevertheless, to address any potential self-selection, I condition on a variety of school-level factors such as geographic zone, treatment group, local infection rate, population density, and number of nearby students attending other treatment schools. As in the previous application, I assume that the idiosyncratic tastes, $\varepsilon_{i,t}$, are logistically distributed in each period.

Table 4 presents IV estimates of $h_{k_1,t} - h_{k_2,t}$ and $J_{k_1\ell} - J_{k_2\ell}$ using data from the deworming experiment. These estimates measure age and gender differences in private payoffs and interaction effects, while controlling for all school-level determinants of treatment uptake. I report four specifications with different combinations of year, student, and school controls.

The results show no significant difference in private payoffs. Yet, I find strong differences in interaction effects, with $J_{kk} > J_{\ell k}$ for $k \neq \ell$, indicating that students prefer to conform to peers of the same type. The degree of own-type conformity, as measured by the parameter $\delta_{k_1 k_2} = J_{k_1 k_1} + J_{k_2 k_2} - J_{k_1 k_2} - J_{k_2 k_1}$, is highest for girls and boys under 14 ($k_1, k_2 \in \{1, 2\}$), implying that gender-based homophily is particularly pronounced among younger students.

The finding that students prefer to conform to similar peers helps to clarify the underlying mechanisms driving spillovers in this setting. In particular, this pattern of peer influence is difficult to reconcile with a pure free-riding motive, which would predict uniform spillovers or spillovers proportional to the epidemiological externality imposed by each type. Rather, it is more consistent with an imitation motive, where peer effects depend on social identity.

This finding also sheds light on the distributional impacts of school-based policies. Within schools, treatment uptake often differs greatly by age and gender, while the magnitude and direction of these gaps vary substantially from one school to another (Appendix Figure A.2). This variation suggests that school-level factors contribute to disparities in treatment uptake. By isolating the contribution of nonuniform peer effects in schools, the generalized interactions framework uncovers how social interactions affect these disparities. My results show that peer effect differences strongly influence which students decide to take treatment. Accounting for these effects can help policymakers target treatment access more effectively.

As before, I conduct three robustness checks. First, in Appendix Table A.8, I test whether the parameters $h_{k,t}$ and $J_{k\ell}$ depend on observed school characteristics, such as the treatment group or local infection rate. I find no evidence of such dependence. Second, in Appendix Figure A.3, I plot the IV estimates across alternative network partitioning rules, and I find that they are relatively stable. Third, in Appendix Table A.9, I validate the finite-sample performance of the estimator using simulations calibrated to the number of schools, school size and demographic composition, and the estimated interaction effects in the deworming data.

program (post-2000), after schools started charging deworming fees, while I study uptake before such fees existed.

TABLE 4
DEWORMING IN SCHOOLS: IV ESTIMATES OF SPILLOVER DIFFERENCES BY AGE AND GENDER

	(1)	(2)	(3)	(4)
<i>Girls ≤ 13 and Boys ≤ 13</i>				
$h_1 - h_2$	0.928 (0.69)	0.857 (0.79)	1.431 (1.67)	1.256 (1.99)
$J_{11} - J_{21}$	8.293 (2.53)	8.679 (2.74)	11.428 (4.38)	12.273 (4.97)
$J_{12} - J_{22}$	-9.382 (2.93)	-10.029 (3.26)	-13.827 (4.25)	-15.598 (5.17)
$J_{13} - J_{23}$	0.071 (0.71)	0.397 (0.86)	0.378 (1.16)	0.600 (1.13)
<i>Girls ≤ 13 and Boys > 13</i>				
$h_1 - h_3$	-0.309 (0.89)	-0.555 (0.97)	0.873 (1.41)	1.266 (1.17)
$J_{11} - J_{31}$	5.551 (2.78)	6.892 (3.02)	7.401 (4.13)	7.824 (3.88)
$J_{12} - J_{32}$	-0.132 (2.38)	-2.382 (2.95)	-5.055 (4.45)	-6.311 (3.91)
$J_{13} - J_{33}$	-4.844 (0.57)	-3.707 (0.93)	-2.633 (1.23)	-2.480 (1.43)
<i>Boys ≤ 13 and Boys > 13</i>				
$h_2 - h_3$	-1.237 (1.09)	-1.413 (0.86)	-0.559 (1.25)	0.010 (1.46)
$J_{21} - J_{31}$	-2.743 (3.22)	-1.787 (3.02)	-4.027 (4.49)	-4.449 (5.38)
$J_{22} - J_{32}$	9.250 (3.06)	7.647 (3.20)	8.772 (5.43)	9.287 (6.51)
$J_{23} - J_{33}$	-4.914 (0.47)	-4.104 (0.61)	-3.012 (1.00)	-3.080 (1.02)
Year Fixed Effects	No	Yes	Yes	Yes
Student Controls	No	No	Yes	Yes
School Controls	No	No	No	Yes
1st Stage $F(\bar{\omega}_n^1)$	14.10	11.20	14.35	13.09
1st Stage $F(\bar{\omega}_n^2)$	12.05	9.97	11.72	10.40
1st Stage $F(\bar{\omega}_n^3)$	35.60	16.69	18.56	17.88

Notes. This table reports estimated differences in spillover effects in treatment uptake, using the estimand in (17). Student controls are prior eligibility, uptake, and attendance. School controls are local infection rate, pop. density, and nearby pupils at other treatment schools. Zone fixed effects are included. Standard errors use the asymptotic variance $Q\Omega Q'$ derived in Appendix B.3, where $\Omega = \text{Var}(Z_n[Y_n^{k_1} - Y_n^{k_2} - X_n'\beta_{k_1, k_2}])$ and $Q = E(Z_n X_n')^{-1}$. For all (k, ℓ) , $h_k - h_\ell$ denotes a time average of $h_{k,t} - h_{\ell,t}$. X_n and Z_n use disjoint 2/3 and 1/3 sample splits with $P = 1$.

5. CONCLUSION

In this paper, I developed new empirical tools to study social interaction effects in discrete choice models in which individuals respond differently to different network members, conforming to some while distinguishing from others. I first analyzed how the signs and magnitudes of endogenous interaction effects influence people's behavior in equilibrium, and the implications for welfare. I then showed how to estimate endogenous interactions from data on individual choices while addressing two identification challenges: unobserved contextual effects and unobserved agent expectations in finite networks. By resolving these challenges, the paper's framework can be readily applied to standard datasets with finite network sizes. I applied the results to data from two large-scale education experiments: Project STAR and the Primary School Deworming Project. In both cases, I found that nonuniform interactions play an important role in shaping people's choices. While this paper provides an initial analysis of nonuniform and mixed-sign interactions in discrete choice models, there is much scope for future work. One valuable extension would be to combine the model with endogenous network formation, allowing individuals to choose their exposure to different peers.

APPENDIX

PROOF OF PROPERTY 1. Define $\mathcal{Q} : [0, 1]^K \rightarrow [0, 1]^K$ and $\mathcal{H} : [0, 1]^K \rightarrow \mathbb{R}^K$ such that:

$$\mathcal{Q}_k(m) = F_{\varepsilon|k}(h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell) \quad \text{and} \quad \mathcal{H}_k(m) = m^k - \mathcal{Q}_k(m), \quad \text{for } k = 1, \dots, K.$$

Since no root of $\mathcal{H}$ lies on the boundary of $[0, 1]^K$, I restrict attention to the interior $(0, 1)^K$.¹ Letting $\mathbf{D}_{\mathcal{H}}$ denote the Jacobian of $\mathcal{H}$, define $\mathcal{C} = \{m \in (0, 1)^K : \det(\mathbf{D}_{\mathcal{H}}(m)) = 0\}$. By Sard's Theorem, $\mathcal{H}(\mathcal{C})$ has Lebesgue measure zero. Also, for any given $y = \mathcal{H}(m)$, almost all $\{F_{\varepsilon|k}\}_k$ ensure that $y \notin \mathcal{H}(\mathcal{C})$. In particular, for $y = \mathbf{0}_K$, it follows that $\mathbf{D}_{\mathcal{H}}(m)$ is almost always nonsingular at every root of $\mathcal{H}$. By the inverse function theorem, each root of $\mathcal{H}$ is isolated. Since the equilibrium set is compact and has only isolated points, it must be finite.

To prove that the number of equilibria is odd, I apply the Poincaré-Hopf Index Theorem (Milnor, 1965, Ch. 6). To do so, first note that the set of equilibria $\mathcal{M}$ lies in $(0, 1)^K$, which is diffeomorphic to an open disk. Also, the vector field $\mathcal{H}$ is smooth and points outward on the boundary, since $\lim_{m^k \rightarrow 0} \mathcal{H}_k(m) < 0$ and $\lim_{m^k \rightarrow 1} \mathcal{H}_k(m) > 0$ for all k . Moreover, for almost all $\{F_{\varepsilon|k}\}_k$, $\mathcal{H}$ has finitely-many isolated roots m , each with $\det(\mathbf{D}_{\mathcal{H}}(m)) \neq 0$. By the index theorem: $\sum_{m \in \mathcal{M}} \text{index}_m(\mathcal{H}) = 1$ where $\text{index}_m(\mathcal{H}) = 2\mathbb{1}\{\det(\mathbf{D}_{\mathcal{H}}(m)) > 0\} - 1$.

Finally, suppose there exist d_s equilibria where $\rho(\mathbf{D}(m)) < 1$. At each of these equilibria:

$$\det(\mathbf{D}_{\mathcal{H}}(m)) = \det(I - \mathbf{D}(m)) = \prod_{k=1}^K (1 - \lambda_k(m)) > 0,$$

where $\{\lambda_k(m)\}_{k=1}^K$ are the eigenvalues of $\mathbf{D}(m)$. By the index theorem, there must also be at least $d_s - 1$ equilibria m at which $\det(\mathbf{D}_{\mathcal{H}}(m)) < 0$, each one satisfying $\rho(\mathbf{D}(m)) > 1$.

Q.E.D.

PROOF OF LEMMA 1. Horn and Johnson (1985), Thm. 5.6.9-5.6.10, show that $\rho(\mathbf{D}(m))$ is the greatest lower bound for the set of all matrix norms of $\mathbf{D}(m)$. In particular, they show:

- (a) $\rho(\mathbf{D}(m)) \leq \|\mathbf{D}(m)\|$ for any matrix norm $\|\cdot\|$.
- (b) For any $\epsilon > 0$, there is a matrix norm $\|\cdot\|$ where $\|\mathbf{D}(m)\| \leq \rho(\mathbf{D}(m)) + \epsilon$.

Let $\rho(\mathbf{D}(m)) < 1$. Setting $\epsilon < 1 - \rho(\mathbf{D}(m))$, it follows from (b) that $\|\mathbf{D}(m)\| < 1$ for some matrix norm $\|\cdot\|$. The system (5) is a contraction map at m under this norm, which means:

$$\|m_t - m\| = \|\mathcal{Q}(m_{t-1}) - \mathcal{Q}(m)\| \leq \kappa \|m_{t-1} - m\|,$$

¹Since each $F_{\varepsilon|k}$ has positive density everywhere, $\mathcal{H}_k(m) < 0$ when $m^k = 0$ and $\mathcal{H}_k(m) > 0$ when $m^k = 1$.

where $\kappa \in [0, 1)$ for any vector m_{t-1} that lies in a sufficiently small neighborhood of m . Iterating on this inequality implies that $\|m_t - m\| \leq \kappa^t \|m_0 - m\|$, where $\lim_{t \rightarrow \infty} \kappa^t = 0$.

Next, let $\rho(\mathbf{D}(m)) > 1$. By (a), $\|\mathbf{D}(m)\| > 1$ for any matrix norm $\|\cdot\|$. By [Henry \(1981\)](#), Thm. 5.1.5, there exists a scalar $u > 0$ such that, for any $\delta > 0$, there is an initial iterate m_0 where $\|m_0 - m\| < \delta$ and where some future iterate m_t , $t \geq 1$, satisfies $\|m_t - m\| \geq u$.

Q.E.D.

PROOF OF PROPERTY 2. First, suppose $\mathbf{D}(\cdot)$ is a nonnegative matrix. Assume there is an equilibrium $m \in (0, 1)^K$ with $\rho(\mathbf{D}(m)) > 1$. The Perron-Frobenius theorem gives a non-zero vector $x \geq 0$ where $\mathbf{D}(m)x = \rho(\mathbf{D}(m))x$. For sufficiently small $\delta > 0$, let $a = m - \delta x$ and $b = m + \delta x$, both in $(0, 1)^K$. Taking the first-order Taylor approximation about m gives:

$$\mathcal{Q}(m \pm \delta x) = m \pm \delta \rho(\mathbf{D}(m))x + o(\delta).$$

Thus, $\mathcal{Q}(a) \leq a$ and $\mathcal{Q}(b) \geq b$, with strict inequalities in coordinates where $x_k > 0$ and equality where $x_k = 0$. These inequalities imply that $\mathcal{Q}$ maps each of the intervals $[0_K, a]$ and $[b, 1_K]$ into itself, as $0_K < \mathcal{Q}(0_K) \leq \mathcal{Q}(a) \leq a$ and $b \leq \mathcal{Q}(b) \leq \mathcal{Q}(1_K) < 1_K$. Brouwer's fixed point theorem yields two equilibria $\underline{m} \leq a$ and $\overline{m} \geq b$, both distinct from m . If either is unstable, repeating the argument guarantees another equilibrium farther downward or upward, respectively. Property 1 implies the equilibrium set is generically finite and $\det(I - \mathbf{D}) \neq 0$ at every equilibrium. Because $\mathbf{D}(m)$ is nonnegative, its spectral radius is an eigenvalue by Perron-Frobenius. Thus, $\det(I - \mathbf{D}(m)) \neq 0$ rules out $\rho(\mathbf{D}(m)) = 1$. Both chains must therefore terminate at distinct equilibria with $\rho(\mathbf{D}) < 1$, which are locally stable by Lemma 1.

I next extend the result to settings satisfying Assumption A. Let $\mathbf{B}$ be an invertible matrix for which $\mathbf{B}\mathbf{D}(\cdot)\mathbf{B}^{-1}$ is nonnegative. For $\mathcal{I} = [0, 1]^K$, define $\hat{\mathcal{Q}}: \mathbf{B}\mathcal{I} \rightarrow \mathbf{B}\mathcal{I}$ such that $\hat{\mathcal{Q}}(m) = \mathbf{B}\mathcal{Q}(\mathbf{B}^{-1}m)$. Its Jacobian is $\mathbf{D}_{\hat{\mathcal{Q}}}(m) = \mathbf{B}\mathbf{D}(\mathbf{B}^{-1}m)\mathbf{B}^{-1}$. Since an invertible linear transformation preserves interior points, $\text{int}(\mathbf{B}\mathcal{I}) = \mathbf{B}\text{int}(\mathcal{I})$. Thus, Assumption A ensures that $\mathbf{D}_{\hat{\mathcal{Q}}}(\cdot) \geq 0$ throughout $\text{int}(\mathbf{B}\mathcal{I})$. It follows that the mapping $\hat{\mathcal{Q}}$ is monotone on $\text{int}(\mathbf{B}\mathcal{I})$.

Since $\hat{\mathcal{Q}}(\mathbf{B}m) = \mathbf{B}\mathcal{Q}(m) = \mathbf{B}m$, the vector m is a fixed point of $\mathcal{Q}$ if and only if $\mathbf{B}m$ is a fixed point of $\hat{\mathcal{Q}}$. The Jacobians $\mathbf{D}(m)$ and $\mathbf{D}_{\hat{\mathcal{Q}}}(\mathbf{B}m)$ are similar, so the corresponding fixed points share the same eigenvalues. Thus, while the ordering of $\mathcal{Q}$ and $\hat{\mathcal{Q}}$'s fixed points could differ, their number and local stability properties are the same. Applying the preceding argument to the monotone map $\hat{\mathcal{Q}}$ establishes that, under Assumption A, an unstable equilibrium implies multiple locally stable equilibria. The converse follows directly from Property 1.

Q.E.D.

PROOF OF PROPERTY 3. If actions take values in the set $\{-1, 1\}$, then $\mathcal{W}_k(m)$ equals:

$$\mathcal{W}_k(m) = \mathbb{E} \left(\left| h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell + \varepsilon_i \right| \right) + \eta_k + \mathbb{E}(\xi_i | k).$$

Since each $F_{\varepsilon|k}$ is symmetric about zero, $\mathcal{W}_k(m)$ is strictly increasing in $|h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell|$. Also, in equilibrium $|\mathbb{E}(\bar{\omega}^k)|$ is strictly increasing in $|h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell|$. Therefore, it must be that $\operatorname{argmax}_{m \in \mathcal{M}} \mathcal{W}_k(m) = \operatorname{argmax}_{m \in \mathcal{M}} |\mathbb{E}(\bar{\omega}^k)|$. The second part of Property 3 follows directly from the observations that $\lim_{h_k \rightarrow \infty} \mathbb{E}(\bar{\omega}^k) = 1$ and $\lim_{h_k \rightarrow -\infty} \mathbb{E}(\bar{\omega}^k) = -1$.

Q.E.D.

PROOF OF PROPERTY 4. Suppose $J_{kj_1} J_{j_1 j_2} \cdots J_{j_M k} \geq 0$ for any types $k, j_1, j_2, \dots, j_M$. By Lemma A.4 of the Online Appendix, there exists a diagonal matrix $\mathbf{B} = \operatorname{diag}[\gamma_1^k, \dots, \gamma_K^k]$, where $\gamma_\ell^k = 2\mathbb{1}\{J_{kj_1} J_{j_1 j_2} \cdots J_{j_M \ell} \geq 0 \text{ for all types } j_1, j_2, \dots, j_M\} - 1$, for which $\mathbf{B}\mathbf{J}\mathbf{B}^{-1}$ is nonnegative. As shown in the proof of Property 2, $\hat{\mathcal{Q}}(m) = \mathbf{B}\mathcal{Q}(\mathbf{B}^{-1}m)$ is monotone on $\operatorname{int}(\mathbf{B}\mathcal{I})$. Since $(\mathbf{B}\mathcal{I}, \leq)$ is a complete lattice, Tarski's fixed point theorem ensures that the set of fixed points of $\hat{\mathcal{Q}}$ is a complete lattice and, consequently, contains both a greatest and least fixed point. If $\mathbf{B}\bar{m}$ is the greatest fixed point of $\hat{\mathcal{Q}}(m)$, then $\bar{m} = \operatorname{argmax}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^\ell)$ if $\gamma_\ell^k = 1$ and $\bar{m} = \operatorname{argmin}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^\ell)$ if $\gamma_\ell^k = -1$. Alternatively, if $\mathbf{B}\underline{m}$ is the least fixed point of $\hat{\mathcal{Q}}(m)$, then $\underline{m} = \operatorname{argmin}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^\ell)$ if $\gamma_\ell^k = 1$ and $\underline{m} = \operatorname{argmax}_{m \in \mathcal{M}} \mathbb{E}(\bar{\omega}^\ell)$ if $\gamma_\ell^k = -1$.

Q.E.D.

PROOF OF THEOREM 1. Fix k_1 and k_2 . Since $F_{\varepsilon|k_1}$ is strictly increasing, it follows that:

$$\begin{aligned} m_n^{k_1} &= F_{\varepsilon|k_1} \left(h_{k_1} - h_{k_2} + \sum_{\ell=1}^K (J_{k_1 \ell} - J_{k_2 \ell}) m_n^\ell + F_{\varepsilon|k_2}^{-1}(m_n^{k_2}) \right) \\ &= F_{\varepsilon|k_1} \left(\overline{h_{k_1} - h_{k_2}} + \sum_{\ell=1}^K (\overline{J_{k_1 \ell} - J_{k_2 \ell}}) m_n^\ell + F_{\varepsilon|k_2}^{-1}(m_n^{k_2}) \right) \end{aligned}$$

if and only if $(h_{k_1} - h_{k_2}) - (\overline{h_{k_1} - h_{k_2}}) = \sum_{\ell=1}^K [(\overline{J_{k_1 \ell} - J_{k_2 \ell}}) - (J_{k_1 \ell} - J_{k_2 \ell})] m_n^\ell$. This property holds for all n . The rank condition in Assumption B.2 implies that $h_{k_1} - h_{k_2} = \overline{h_{k_1} - h_{k_2}}$ and $J_{k_1 \ell} - J_{k_2 \ell} = \overline{J_{k_1 \ell} - J_{k_2 \ell}}$ for any type $\ell \in \{1, \dots, K\}$. Since k_1 and k_2 are chosen arbitrarily, this result extends to all $k_1, k_2 \in \{1, \dots, K\}$. The proof in the $\alpha_n = W_n' d$ case is analogous.

Q.E.D.

PROOF OF THEOREM 2. I demonstrate that $\{F_{\varepsilon|k}\}_k$ and c are identified up-to-scale, and the remainder follows as in the proof of Theorem 1. Fix any type k . By Assumption B.3, there is some element x_j of X with continuous variation on $\mathbb{R}$. Without loss of generality, let $x_j = x_1$, and normalize the coefficient c_1 to one. Also, fix some network n , and define

$\zeta_n^k = h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell} m_n^\ell$. For any agent i in type k and network n with covariates X_i , the expected choice $E(\omega_i | X_i, k, \alpha_n, \{m_n^\ell\}_\ell)$ equals $F_{\varepsilon|k}(\zeta_n^k + X_i' c)$. So, to recover $(c, F_{\varepsilon|k})$, I must show that $F_{\varepsilon|k}(\zeta_n^k + X_i' c) = \bar{F}_{\varepsilon|k}(\zeta_n^k + X_i' \bar{c})$ implies $c = \bar{c}$ and $F_{\varepsilon|k} = \bar{F}_{\varepsilon|k}$ for any $X_i \in \text{supp}(X|k)$. This property is shown by Corollary 5 of Proposition 2 in [Manski \(1988\)](#). Also, since this holds for all k , I thus conclude that c and $\{F_{\varepsilon|k}\}_k$ are identified up-to-scale.

Q.E.D.

PROOF OF THEOREM 3. I prove part (ii) only, as the proof of part (i) is analogous. As a first step, note that for any pair of types k and ℓ : (1) $E(\bar{\omega}_{n,j}^k) = E[E(m_n^k + u_{n,j}^k | n)] = E(m_n^k)$ for each $j \in \{a, b\}$; (2) $E(\bar{\omega}_{n,a}^k \bar{\omega}_{n,b}^\ell) = E[E((m_n^k + u_{n,a}^k)(m_n^\ell + u_{n,b}^\ell) | n)] = E(m_n^k m_n^\ell)$; and (3) $E(\bar{\omega}_{n,b}^\ell F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k)) = E[E((m_n^\ell + u_{n,b}^\ell) F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k) | n)] = E(m_n^\ell F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k))$. Hence, I can write:

$$\begin{aligned} E(Z_n X_n')^{-1} E(Z_n F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k)) &= E(X_n^* (X_n^*)')^{-1} E(Z_n F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k)) && \text{by (1) and (2)} \\ &= E(X_n^* (X_n^*)')^{-1} E(X_n^* F_{\varepsilon|k}^{-1}(\bar{\omega}_{n,a}^k)) && \text{by (3)} \\ &= \underbrace{E(X_n^* (X_n^*)')^{-1} E(X_n^* F_{\varepsilon|k}^{-1}(m_n^k))}_{=\beta_k} + E(X_n^* (X_n^*)')^{-1} E(X_n^* v_{n,a}^k), \end{aligned}$$

where $X_n^* = (1, W_n', m_n^1, \dots, m_n^K)'$. As a notational shorthand, define $q_k = F_{\varepsilon|k}^{-1}$. Expanding $q_k(\bar{\omega}_{n,a}^k)$ about m_n^k through order $2P - 1$, for $P \geq 2$, while retaining the remainder, gives:²

$$\begin{aligned} E(v_{n,a}^k | n) &= E(q_k(\bar{\omega}_{n,a}^k) - q_k(m_n^k) | n) \\ &= E\left(\sum_{r=0}^{2P-1} \frac{q_k^{(r)}(m_n^k)}{r!} (\bar{\omega}_{n,a}^k - m_n^k)^r + \rho_{n,a,P}^k - q_k(m_n^k) \mid n\right) \\ &= q_k'(m_n^k) E(\bar{\omega}_{n,a}^k - m_n^k | n) + \sum_{r=2}^{2P-1} \frac{q_k^{(r)}(m_n^k)}{r!} E((\bar{\omega}_{n,a}^k - m_n^k)^r | n) + E(\rho_{n,a,P}^k | n) \\ &= \sum_{r=2}^{2P-1} \frac{q_k^{(r)}(m_n^k)}{r!} B_r\left(0, \frac{\kappa_2(\omega|k, n)}{I_{n,a}^k}, \frac{\kappa_3(\omega|k, n)}{(I_{n,a}^k)^2}, \dots, \frac{\kappa_r(\omega|k, n)}{(I_{n,a}^k)^{r-1}}\right) + E(\rho_{n,a,P}^k | n), \end{aligned}$$

where B_r is the r th complete Bell polynomial, and $\{\kappa_i(\omega|k, n)\}_i$ are cumulants of ω_i given k and n . Each $B_r(0, \frac{\kappa_2(\omega|k, n)}{I_{n,a}^k}, \frac{\kappa_3(\omega|k, n)}{(I_{n,a}^k)^2}, \dots, \frac{\kappa_r(\omega|k, n)}{(I_{n,a}^k)^{r-1}})$ is therefore a polynomial in $(I_{n,a}^k)^{-1}$ with no constant term and coefficients depending only on cumulants of ω_i given k and n . It follows that $E(v_{n,a}^k | n) = \sum_{s=1}^{P-1} a_{n,s}^k / (I_{n,a}^k)^s + r_{n,a,P}^k$, where each coefficient $a_{n,s}^k$ does not depend on the subgroup size $I_{n,a}^k$, and the remainder $r_{n,a,P}^k$ collects all the other terms. Noting that

²For notational simplicity, this proof suppresses the treatment of subsample proportions $\bar{\omega}_{n,a}^k$ equal to zero or one, at which the inverse choice function could be infinite. Online Appendix B.2 specifies an endpoint smoothing convention and verifies the conditions under which the bias expansion and limiting correction below remain valid.

$E(X_n^* v_{n,a}^k) = E(X_n^* E(v_{n,a}^k | n))$ and using independence of the subgroup sizes from X_n^* , write:

$$\begin{aligned} E(Z_n X_n')^{-1} E(Z_n q_k(\bar{\omega}_{n,a}^k)) &= \beta_k + E(X_n^* (X_n^*)')^{-1} E\left(X_n^* \left[\sum_{s=1}^{P-1} \frac{a_{n,s}^k}{(I_{n,a}^k)^s} + r_{n,a,P}^k \right]\right) \\ &= \beta_k + \sum_{s=1}^{P-1} C_{k,s} E((I_{n,a}^k)^{-s}) + E(X_n^* (X_n^*)')^{-1} E(X_n^* r_{n,a,P}^k), \end{aligned}$$

where $C_{k,s} = E(X_n^* (X_n^*)')^{-1} E(X_n^* a_{n,s}^k)$ are constant vectors that do not depend on $\{I_{n,a}^k\}_n$. By the definition of $\mathbf{E}_{k,P}$, both $\sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} = 1$ and $\sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} E[(I_{n,\kappa(p,n)}^k)^{-s}] = 0$ for $s = 1, \dots, P-1$. Therefore, letting $R_{k,P,p} = E(X_n^* (X_n^*)')^{-1} E[X_n^* r_{n,\kappa(p,n),P}^k]$, I obtain:

$$\begin{aligned} E(Z_n X_n')^{-1} E\left[Z_n \sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} q_k(\bar{\omega}_{n,\kappa(p,n)}^k)\right] &= \sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} \left[\beta_k + \sum_{s=1}^{P-1} C_{k,s} E[(I_{n,\kappa(p,n)}^k)^{-s}] + R_{k,P,p} \right] \\ &= \beta_k + \sum_{p=1}^P [\mathbf{E}_{k,P}^{-1}]_{p,1} R_{k,P,p}. \end{aligned}$$

Hence, the P -rule correction eliminates the first $P-1$ inverse-size terms in the expansion. To establish exact identification, it remains to justify the limiting correction as $P \rightarrow \infty$. Online Appendix B.2 gives examples of sequences of partitioning rules for which $\mathbf{E}_{k,P}$ is nonsingular at all P and the resulting Y_n^k is integrable with $E(Y_n^k | m_n, W_n) = q_k(m_n^k)$, so:

$$\beta_k^{IV} = E(X_n^* (X_n^*)')^{-1} E(X_n^* Y_n^k) = E(X_n^* (X_n^*)')^{-1} E(X_n^* q_k(m_n^k)) = \beta_k.$$

Q.E.D.

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